Token and positional embedding → Attention Mechanism → Feed Forward Network → Repeat → Final Output

An aardvark eats ... *Input text (length 3)*

Tokenize (some words get split into smaller tokens)

Token IDs (100,000 unique IDs)
01516 00001 12791

Look up embedding for each token

Embedding matrix (100,000 × 512)

$$\begin{array}{r} \text{a } 00000 \\ \text{aardvark } 00001 \\ \text{ab } 00002 \\ \cdots \\ \text{zyzzyva } 99999 \end{array} \begin{bmatrix} 0.1 & 0.3 & -0.2 & \cdots \\ -0.4 & 0.5 & 0.9 & \cdots \\ 0.8 & -0.6 & 0.3 & \cdots \\ \cdots & & & \\ -0.1 & -0.8 & 0.6 & \cdots \end{bmatrix}$$

51.2M parameters

Token embedding (3 × 512)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 0.9 & 0.7 & -0.1 & \cdots \\ -0.4 & 0.5 & 0.9 & \cdots \\ -0.2 & -0.3 & 0.1 & \cdots \end{bmatrix}$$

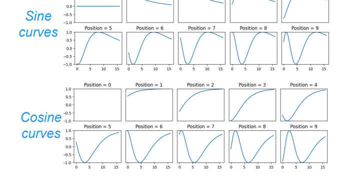
Add token and position embeddings +

Make the position embedding matrix
- Fill 3 rows with 256 sine and 256 cosine wave values
- The wavelengths "stretch out" as you move from column 1 to 512
- Each new row has a shorter starting wavelength and slight phase shift
- Unique: each position (row) has a unique set of values
- Relative: Small phase shifts allow different columns to indicate relative distance between every pair of words

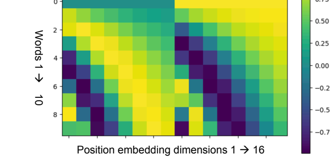
Position embedding (3 × 512)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 0 & 1 & \cdots & 0 & 1 \\ 0.84 & 0.54 & \cdots & 0.11 & 0.99 \\ 0.91 & -0.42 & \cdots & 0.21 & 0.98 \end{bmatrix}$$

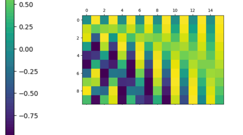
Example with 10 rows, 16 dimensions



Heatmap of example position matrix



In practice, sine/cosine are interleaved



Multi-Head Attention input (3 × 512)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 0.90 & 1.70 & -0.10 & \cdots \\ 0.04 & 1.04 & 1.40 & \cdots \\ 0.71 & -0.72 & 1.06 & \cdots \end{bmatrix}$$

Multiply by Query, Key, and Value weight matrices
- Each column takes a weighted sum of the embedding dimensions for one token
- Each token gets 512 different weighted sums of its 512 embedding values
- Repeat for Query, Key, and Value weights

× *"Query" weight matrix (512 × 512)*

$$\begin{bmatrix} 0.85 & 0.12 & 0.09 & \cdots \\ -0.72 & -0.75 & -0.03 & \cdots \\ -0.61 & -0.99 & -0.39 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

262k parameters

× *"Key" weight matrix (512 × 512)*

$$\begin{bmatrix} 0.45 & 0.93 & -0.78 & \cdots \\ -0.51 & -0.65 & -0.16 & \cdots \\ -0.06 & -0.12 & 0.81 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

262k parameters

× *"Value" weight matrix (512 × 512)*

$$\begin{bmatrix} -0.48 & 0.96 & -0.29 & \cdots \\ -0.04 & 0.52 & 0.83 & \cdots \\ -0.65 & -0.62 & -0.42 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

262k parameters

Query, Key, Value matrices still have one row per token

The dimensions are different learned weighted combinations of the original embedding dimensions

"Query" matrix (3 × 512)

$$\begin{bmatrix} -0.50 & -0.12 & 0.73 & \cdots \\ 0.72 & 0.65 & -0.25 & \cdots \\ -0.41 & 0.90 & -0.93 & \cdots \end{bmatrix}$$

"Key" matrix (3 × 512)

$$\begin{bmatrix} 0.77 & -0.30 & -0.37 & \cdots \\ 0.98 & -0.01 & -0.55 & \cdots \\ 0.11 & -0.66 & 0.60 & \cdots \end{bmatrix}$$

"Value" matrix (3 × 512)

$$\begin{bmatrix} 0.99 & -0.20 & 0.72 & \cdots \\ -0.98 & 0.54 & -0.95 & \cdots \\ 0.10 & -0.45 & -0.39 & \cdots \end{bmatrix}$$

Split Q, K, V matrices width-wise into 8 pieces

These will perform calculations for 8 distinct attention "heads"

Split → *8 mini-Query matrices (3 × 64)*

$$\begin{bmatrix} -0.50 & -0.12 & 0.73 & \cdots \\ 0.72 & 0.65 & -0.25 & \cdots \\ -0.41 & 0.90 & -0.93 & \cdots \end{bmatrix}$$

Split → *8 mini-Key matrices (3 × 64)*

$$\begin{bmatrix} 0.77 & -0.30 & -0.37 & \cdots \\ 0.98 & -0.01 & -0.55 & \cdots \\ 0.11 & -0.66 & 0.60 & \cdots \end{bmatrix}$$

Split → *8 mini-Value matrices (3 × 64)*

$$\begin{bmatrix} 0.99 & -0.20 & 0.72 & \cdots \\ -0.98 & 0.54 & -0.95 & \cdots \\ -0.10 & -0.45 & -0.39 & \cdots \end{bmatrix}$$

Residual connection

$$\underbrace{\overset{\text{Query \#1 } (3\times 64)}{\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} -0.50 & -0.12 & 0.73 & \cdots \\ 0.72 & 0.65 & -0.25 & \cdots \\ -0.41 & 0.90 & -0.93 & \cdots \end{bmatrix}} \times \overset{\text{Key \#1 transpose } (64\times 3)}{\begin{array}{ccc} \text{an} & \text{aardvark} & \text{eats} \\ \hline 0.77 & 0.98 & 0.11 \\ -0.30 & -0.01 & -0.66 \\ -0.37 & -0.55 & 0.60 \\ \vdots & \vdots & \vdots \end{array}} = \overset{\text{Attention scores matrix \#1 } (3\times 3)}{\begin{array}{r|ccc} & \text{an} & \text{aardvark} & \text{eats} \\ \hline \text{an} & -0.62 & -0.89 & 0.46 \\ \text{aardvark} & 0.45 & 0.84 & -0.50 \\ \text{eats} & -0.24 & 0.10 & -1.12 \end{array}} \xrightarrow{\text{Softmax rows}} \overset{\text{Attention weights matrix \#1 } (3\times 3)}{\begin{array}{r|ccc} & \text{an} & \text{aardvark} & \text{eats} \\ \hline \text{an} & 0.21 & 0.16 & 0.63 \\ \text{aardvark} & 0.35 & 0.51 & 0.14 \\ \text{eats} & 0.36 & 0.50 & 0.14 \end{array}} \times \overset{\text{"Value" matrix } (3\times 64)}{\begin{bmatrix} 0.99 & -0.20 & 0.72 & \cdots \\ -0.98 & 0.54 & -0.95 & \cdots \\ 0.10 & -0.45 & -0.39 & \cdots \end{bmatrix}} = \overset{\text{Context matrix \#1 } (3\times 64)}{\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 0.11 & -0.24 & -0.25 & \cdots \\ -0.14 & 0.15 & -0.29 & \cdots \\ -0.13 & 0.14 & -0.27 & \cdots \end{bmatrix}}}_{\text{Attention head \#1}}$$

Repeat for all 8 heads

8 Context matrices (3× 64 each)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 0.11 & -0.24 & -0.25 & \cdots \\ -0.14 & 0.15 & -0.29 & \cdots \\ -0.13 & 0.14 & -0.27 & \cdots \end{bmatrix}$$

Concatenate

Concatenated matrix (3× 512)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 0.11 & -0.24 & -0.25 & \cdots \\ -0.14 & 0.15 & -0.29 & \cdots \\ -0.13 & 0.14 & -0.27 & \cdots \end{bmatrix}$$

× *Attention output projection weight matrix (512 × 512)*

$$\begin{bmatrix} 0.13 & -0.23 & 0.65 & \cdots \\ 0.05 & -0.93 & -0.81 & \cdots \\ 0.76 & -0.52 & 0.92 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

262k parameters

= *Multi-Head Attention output (3 × 512)*

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} -0.18 & 0.32 & -0.04 & \cdots \\ -0.23 & 0.05 & -0.48 & \cdots \\ -0.22 & 0.04 & -0.45 & \cdots \end{bmatrix}$$

+ *Multi-Head Attention input (3 × 512)*

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 0.90 & 1.70 & -0.10 & \cdots \\ 0.04 & 1.04 & 1.40 & \cdots \\ 0.71 & -0.72 & 1.06 & \cdots \end{bmatrix}$$

Residual connection output matrix (3 × 512)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 0.72 & 2.02 & -0.14 & \cdots \\ -0.19 & 1.09 & 0.92 & \cdots \\ 0.49 & -0.68 & 0.61 & \cdots \end{bmatrix}$$

Normalize each row to mean=0, variance=1

Normalized output matrix (3 × 512)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} -0.13 & 1.06 & -0.93 & \cdots \\ -1.14 & 0.69 & 0.45 & \cdots \\ 0.49 & -1.15 & 0.66 & \cdots \end{bmatrix}$$

Scale and shift all values by learned parameters γ and β.
$y_i = \gamma\ x_i + \beta$
For example, $y_i = 0.9 + 0.2$

2 parameters

Residual connection

Attention output matrix (3 × 512)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 0.08 & 1.15 & -0.63 & \cdots \\ -0.83 & 0.83 & 0.61 & \cdots \\ 0.64 & -0.84 & 0.79 & \cdots \end{bmatrix}$$

× *Feed-forward layer 1 weight matrix (512 × 2048)*

$$\begin{bmatrix} -0.03 & -0.79 & 0.36 & \cdots \\ 0.23 & 0.85 & -0.50 & \cdots \\ 0.40 & 0.11 & -0.78 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

1M parameters

+ *Feed-forward layer 1 bias (3 × 512)*

$$\begin{bmatrix} -0.10 & 0.03 & 0 & \cdots \\ -0.10 & 0.03 & 0 & \cdots \\ -0.10 & 0.03 & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

2048 parameters

Feed-forward Layer 1 matrix (3 × 2048)

$$\begin{bmatrix} -0.09 & 0.88 & -0.06 & \cdots \\ 0.36 & 1.45 & -1.18 & \cdots \\ -0.01 & -1.10 & 0.03 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

ReLU activation (negatives become zero)

Layer 1 activation matrix (3 × 2048)

$$\begin{bmatrix} 0 & 0.88 & 0 & \cdots \\ 0.36 & 1.45 & 0 & \cdots \\ 0 & 0 & 0.03 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

× *Feed-forward layer 2 weight matrix (2048 × 512)*

$$\begin{bmatrix} -0.60 & 0.87 & 0.12 & \cdots \\ 0.96 & -0.67 & 0.39 & \cdots \\ -0.96 & 0.74 & -0.83 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

1M parameters

+ *Feed-forward layer 2 bias (3 × 512)*

$$\begin{bmatrix} 0.4 & -0.19 & 0.2 & \cdots \\ 0.4 & -0.19 & 0.2 & \cdots \\ 0.4 & -0.19 & 0.2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

512 parameters

Feed-forward network output matrix (3 × 512)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 1.24 & -0.78 & 0.54 & \cdots \\ 1.58 & -0.85 & 0.91 & \cdots \\ 0.37 & -0.16 & -0.18 & \cdots \end{bmatrix}$$

+ *Attention output matrix (3 × 512)*

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 0.08 & 1.15 & -0.63 & \cdots \\ -0.83 & 0.83 & 0.61 & \cdots \\ 0.64 & -0.84 & 0.79 & \cdots \end{bmatrix}$$

Residual connection output matrix (3 × 512)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 1.32 & 0.38 & -0.09 & \cdots \\ 0.75 & -0.03 & 1.42 & \cdots \\ 1.01 & -0.99 & 0.97 & \cdots \end{bmatrix}$$

Normalize each row to mean=0, variance=1

Normalized matrix (3 × 512)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 1.09 & -0.22 & -0.87 & \cdots \\ 0.05 & -1.02 & 0.97 & \cdots \\ 0.60 & -1.15 & 0.56 & \cdots \end{bmatrix}$$

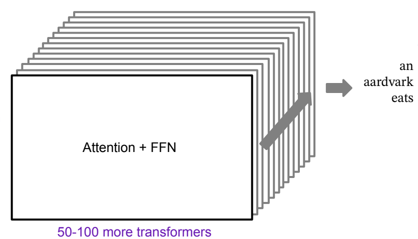
Scale and shift all values by learned parameters γ and β.
$y_i = \gamma\ x_i + \beta$
For example, $y_i = 0.8x - 0.04$

2 parameters

Final output matrix (3 × 512)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 0.83 & -0.22 & -0.74 & \cdots \\ 0.00 & -0.86 & 0.74 & \cdots \\ 0.44 & -0.96 & 0.41 & \cdots \end{bmatrix}$$

Input directly to next attention mechanism



50-100 more transformers

Final output matrix (3 × 512)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} 0.92 & 0.99 & -0.80 & \cdots \\ -0.81 & 0.38 & 0.95 & \cdots \\ 0.49 & -0.78 & 0.44 & \cdots \end{bmatrix}$$

× *Vocab projection matrix (512 × 100,000)*

$$\begin{bmatrix} -0.30 & 0.55 & 0.82 & \cdots \\ 0.21 & -0.06 & 0.19 & \cdots \\ -0.51 & 0.34 & -0.43 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

512k parameters

Vocab probability scores (3 × 100,000)

$$\begin{array}{r} \text{an} \\ \text{aardvark} \\ \text{eats} \end{array} \begin{bmatrix} -0.01 & 3.43 & -3.58 & \cdots \\ -4.00 & -9.51 & -2.55 & \cdots \\ -1.02 & -6.32 & -3.23 & \cdots \end{bmatrix}$$

Softmax to get probabilities (optionally divide all by Temperature T first to smooth or sharpen the probabilities)

Next word probability scores (3 × 100,000)

$$\begin{array}{r|ccccccccc} & \text{a} & \text{aardvark} & \text{ab} & \ldots & \text{ants} & \ldots & \text{zyzzyva} \\ \hline \text{an} & 0.00 & 0.97 & 0.00 & \cdots & 0.00 & \cdots & 0.00 \\ \text{aardvark} & 0.00 & 0.00 & 0.02 & \cdots & 0.00 & \cdots & 0.00 \\ \text{eats} & \boxed{0.005} & 0.000 & 0.00 & \cdots & \boxed{0.99} & \cdots & \boxed{0.001} \end{array}$$

Sample the next word from the last row probabilities

"ants"